Quiz 1: Probability Essentials and Random Variables

1. A dolphin-watching company gives the following guarantee: 'Dolphins seen on every trip — or your money back!' The company wants to investigate the chance of seeing a dolphin on a trip. Which one of the following is a suitable sample space for their investigation?
   (a) $\Omega = \{\text{dolphin sightings}\}$
   (b) $\Omega = \{\text{dolphin-watching trips}\}$
   (c) $\Omega = \{\text{trips on which dolphins are seen}\}$
   (d) $\Omega = \{\text{customers}\}$

2. Some investigators want to know whether married couples are likely to share the same star sign. They select married couples at random and record their star signs. There are 12 possible star signs. The following statements are about the probability formulation for this investigation. Which one of the following statements is **not** correct?
   (a) A suitable sample space would be $\Omega = \{\text{married couples}\}$.
   (b) A suitable sample space would be $\Omega = \{\text{the 12 star signs}\}$.
   (c) A relevant event is $S = \{\text{couple share a star sign}\}$.
   (d) If S is defined as above, the investigators want to know whether or not $P(S) = 1/12$.

3. Consider the random experiment 'pick a person in this class at random'. The sample space is $\Omega = \{\text{people in this class}\}$. Which one of the following statements is **not** an example of an event?
   (a) $H = \{\text{height of the person is less than 170cm}\}$.
   (b) $D = \{\text{person owns a dog}\}$.
   (c) $T = \{\text{person's favourite actor is Tom Cruise}\}$.
   (d) $P = \{\text{person's parents}\}$.

4. Consider the random experiment 'pick a person in this class at random' with sample space $\Omega = \{\text{people in this class}\}$. Define events $F = \{\text{person is female}\}$ and $T = \{\text{person is twenty years old}\}$. Suppose I pick a 19-year-old female. The statements J, K, L, M, and N below describe possible events on Ω. Which combination of the events has occurred?

   J. $\overline{T} \cup F$
   K. $T \cup \overline{F}$
   L. $T \cup F$
   M. $\overline{T} \cap F$
   N. $T \cap \overline{F}$

   (a) J, K only.
   (b) J, K, N only.
   (c) J, K, L, N only.
   (d) M, N only.

5. Suppose events A and B are defined on a sample space Ω, and $P(A) = 0.6$, $P(B) = 0.2$. Which combination of the following statements is true?

   J. We need additional information to be able to calculate $P(\overline{A})$.
   K. We need additional information to be able to calculate $P(A \cup B)$.
   L. We need additional information to be able to calculate $P(A \cap B)$.
   M. The largest possible value of $P(A \cup B)$ is 0.8.
   N. The largest possible value of $P(A \cap B)$ is 0.6.

   (a) K, L, M, N only.
   (b) J, K, L only.
   (c) M only.
   (d) K, L, M only.

6. Let A be an event on a sample space Ω. Some of the statements below have correct notation, and some do not. Which combination of statements is written in **correct notation**?

   J. $P(A \cup \overline{A}) = 1$
   K. $P(A) \cup P(\overline{A}) = 1$
   L. $P(A) + P(\overline{A}) = 1$
   M. $P(A) \cup P(\overline{A}) = \Omega$
   N. $P(A \cup \overline{A}) = P(\Omega)$

   (a) J, L, N only.
   (b) J, L only.
   (c) K only.
   (d) M only.

7. Let X be a random variable. Some of the statements below have correct notation, and some do not. Which combination of statements is written in **correct notation**?

   J. $P(X = 2 \cup 3)$
   K. $P(X = 2 \cup X = 3)$
   L. $P(X = 2) + P(X = 3)$
   M. $\{X = 2\} + \{X = 3\}$
   N. $P(X = 2) \cup P(X = 3)$

   (a) K, L only.
   (b) K, L, M only.
   (c) J, K, N only.
   (d) J, M, N only.

8. Let X be a random variable with the following probability function:

x	1	2	3
$P(X = x)$	0.4	0.5	0.1

   Let $F_X(x)$ be the cumulative distribution function of X. Which one of the following statements is **false**?

   (a) $F_X(2) = 0.9$.
   (b) $F_X(4) = 1$.
   (c) The correct notation for the probability function is $f_x(X)$.
   (d) $P(X < 2) = 0.4$.

9. Let $X \sim \text{Binomial}(5, p)$. This means that X is a random variable counting the number of successes out of 5 independent trials, each of which has probability p of success. Denote a trial by S if it is a success, and by F if it is a failure. We are interested in the outcome $SFFSS$, which indicates that the first trial is a success, the second trial is a failure, the third trial is a failure, and so on. Which one of the following statements is **true**?
   (a) $P(SFFSS) = P(X = 3)$
   (b) $P(SFFSS) = \binom{5}{3} p^3(1-p)^2$
   (c) $P(SFFSS) = p^3$
   (d) $P(SFFSS) = p^3(1-p)^2$

10. Which one of the following random variables X is **not** suitable to be modeled by a Binomial distribution?
   (a) X is the number of heads in 10 tosses of a fair coin.
   (b) X is the number of attempts to start a faulty car before it starts.
   (c) X is the number of girls in a two-child family.
   (d) X is the number of times you get an even number in 10 rolls of a fair die.

11. $P(A \mid B)$ is read as, 'probability of A, given B'. A suitable replacement for the word *given* would be:
   (a) caused by
   (b) including
   (c) within
   (d) over

12. Some doctors are conducting an experiment to trial a vaccine for swine flu. They will take a group of people, vaccinate some of them, and then monitor all the people for a year to see whether they get swine flu. The sample space is $\Omega = \{\text{people in the experiment}\}$. Define events

$$V = \{\text{person is vaccinated}\}, \quad S = \{\text{person gets swine flu}\}.$$

   Which two probabilities will the doctors want to compare at the end of the experiment?
   (a) $P(S \mid V)$ and $P(S \mid \overline{V})$.
   (b) $P(S \mid V)$ and $P(\overline{S} \mid V)$.
   (c) $P(V \mid S)$ and $P(V \mid \overline{S})$.
   (d) $P(S \cap V)$ and $P(S \cap \overline{V})$.

13. The doctors from Q12 further want to investigate whether the vaccine has a different effect on older people and younger people. Define further events

$$O = \{\text{person is over 60 years old}\}, \quad Y = \{\text{person is under 30 years old}\}.$$

   Which two probabilities should the doctors compare for this investigation?
   (a) $P(S \mid V \cap O)$ and $P(S \mid V \cap Y)$.
   (b) $P(S \cap O \mid V)$ and $P(S \cap Y \mid V)$.
   (c) $P(S \cap V \mid O)$ and $P(S \cap V \mid Y)$.
   (d) $P(S \mid O)$ and $P(S \mid Y)$.